

WEEK-3 ASSESSMENT

1. Which of the following statements is true about kernel and memory (RAM) in linux and xv6?

1 Point

- ☒ Kernel resides in the upper part of the memory
- ☐ Kernel resides in the lower part of the memory
- ☐ Kernel does not reside completely in the memory
- ☐ Kernel resides anywhere in memory and this is kept track by a universal page table

2. State True or false: There is a Kernel stack and a User stack in the process address space to ensure that the attacks that target the stack will not affect the kernel.

- ☒ True
- ☐ False

3. A process after running for a while is preempted from the processor for other processes to run. Then it's process state is moved to ____

- ☐ NEW
- ☒ READY
- ☐ RUNNING
- ☐ BLOCK

4. What is the output in the child process?

```
int main()
{
    Char fg = 'O';
    int pid = fork();
    if(pid != 0)
    {
        printf("fg = %d", fg+2)
    }
    else
    {
        printf("fg = %d", fg-2);
    }
}
```

☐ fg = M

☐ fg = Q

☒ fg = 77

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5. Which of the following are true about fork() system call (Multiple answers are correct)?

1 Point

- ☒ All parent pages are initially marked as shared
- ☐ fork() returns 0 to the parent process
- ☐ fork() returns parent's process id to the child process
- ☒ To execute another program in the child process, another system call exec is evoked in the child process after fork

6. Which of the methods can we use to terminate a process involuntarily?

1 Point

- ☐ fork
- ☒ Kill
- ☐ exec

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7. Which of the following are true about zombie entries?

- ☒ A zombie entry is removed when its parent reads its exit status
- ☐ When the process terminates, it becomes a zombie and is immediately removed
- ☐ All zombies are a resource leak and will continue to exist infinitely
- ☐ If a process terminates creating at least one intentional orphan, it does not become a zombie

8. State True or False: When an exit system call gets executed, a wakeup signal is sent to its parent process because the parent may be sleeping due to a wait.

- ☒ True
- ☐ False

9. State True or False: When the wait system call is invoked in the parent, it's going to be blocked until all of its child processes exit.

- ☐ True
- ☒ False

10. Which of the following is not part of an ELF header?

- ☐ Magic number
- ☐ Type
- ☒ Network details
- ☐ Machine details